

Rigorous Asymptotics of Orthogonal Polynomials, Structured Matrices, and Combinatorial Sequences

A Synthesis of Spectral Theory and Computational Limits

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July 2026

The interplay between continuous functional analysis and discrete algebraic structures forms the bedrock of modern approximation theory, spectral analysis, and numerical linear algebra. The transition from abstract measures to discrete sequences—whether through Fourier coefficients, moments, or orthogonal polynomial recurrences—yields highly structured linear operators, most notably Hankel and Toeplitz matrices. The asymptotic behaviors of these structures dictate the stability of numerical algorithms, the limiting distributions of probabilistic models, and the spectral properties of physical systems. Following the strict laws of inference and the absolute requirement for rigorous proof in all mathematical claims, this treatise provides an exhaustive examination of the classical moment problem, the Szegő limit theorems, the conditioning of structured matrices, and the asymptotic expansions of associated combinatorial sequences. Furthermore, purported meta-computational models of reality are subjected to a rigorous mathematical audit to distinguish heuristically derived philosophies from proven theorems. The objective is to pull back and observe the overarching architecture of discrete-to-continuous limits, ensuring that every numerical result and mathematical claim is strictly audited.

The Classical Moment Problem and Hankel Matrix Positivity

The classical moment problem asks a deceptively simple question: under what conditions can a given sequence of real numbers $(s_n)_{n=0}^{\infty}$ be represented as the exact moments of a positive measure μ supported on a specific subset of the real line? The problem bifurcates into three distinct topological cases: the Hamburger moment problem (support on the entire real line $\mathbb{R}$), the Stieltjes moment problem (support on the half-line $[0, \infty)$), and the Hausdorff moment problem (support on a bounded interval, typically $[0,1]$ or $[-1,1]$).

The fundamental theorem of existence for the Hamburger moment problem dictates that the sequence (s_n) corresponds to a positive measure if and only if the associated infinite Hankel matrix H , defined by its entries $H_{i,j} = s_{i+j}$ for $i, j \geq 0$, is positive semidefinite. The proof of necessity is straightforwardly derived from the linearity of the integral operator and the algebraic non-negativity of the square of any polynomial with real coefficients. Let $P(x) = \sum_{i=0}^n c_i x^i$ be an arbitrary polynomial. The expectation of its square with respect to a positive measure μ must be non-negative:

$$\int \left(\sum_{i=0}^n c_i x^i \right)^2 d\mu(x) = \sum_{i=0}^n \sum_{j=0}^n c_i c_j \int x^{i+j} d\mu(x) = \sum_{i,j=0}^n c_i c_j s_{i+j} \geq 0$$

Because the rightmost expression is exactly the quadratic form $c^T H_n c$ for the truncated $(n+1) \times (n+1)$ Hankel matrix H_n , the matrix must be positive semidefinite. By Haviland's theorem, this positive semidefiniteness is also sufficient. Haviland's proof relies on the Riesz representation theorem and the property that in the univariate case, every non-negative polynomial can be factored as a sum of squares. Therefore, a linear functional that is positive on sums of squares is positive on all non-negative polynomials, allowing it to be extended to continuous functions with compact support, thereby guaranteeing the existence of a representing measure.

If such a measure exists, the question of uniqueness—determinacy versus indeterminacy—arises. The determinacy of a moment problem is fundamentally linked to the growth rate of the sequence (s_n) . If the moments grow too rapidly, the degrees of freedom in the solution space expand into infinite dimensions, leading to uncountably many non-equivalent measures possessing the identical moment sequence. Torsten Carleman provided a rigorous sufficient criterion for determinacy in the Hamburger case. Carleman's condition states that if the series involving the reciprocal roots of the even-order moments diverges,

$$\sum_{n=1}^{\infty} (s_{2n})^{-\frac{1}{2n}} = \infty$$

then the moment problem possesses a unique solution. The proof of Carleman's condition operates on the principle that the divergence of this series restricts the asymptotic tail of the distribution, forcing it to decay sufficiently fast. If the tail were allowed to contain too much mass (as in the case of the log-normal distribution, which is indeterminate), the moments would grow fast enough for the series to converge.

Alternatively, determinacy can be evaluated through the recurrence coefficients of the associated orthonormal polynomials. If the series of the reciprocal of the off-diagonal Jacobi coefficients $\sum b_n^{-1}$ diverges, the measure is uniquely determined. Furthermore, modern spectral analysis proves that a moment problem is determinate if and only if the smallest eigenvalue of the Hankel matrix $H_n = (s_{i+j})_{i,j=0}^n$ tends to zero as $n \rightarrow \infty$. This eigenvalue decay serves as a critical bridge between the theoretical geometry of the moment space and the numerical stability of Hankel systems.

Orthogonal Polynomials and Recurrence Dynamics

Given a positive measure μ with finite moments, one can construct a sequence of monic orthogonal polynomials $P_n(x)$ by applying the Gram-Schmidt orthogonalization process to the standard monomial basis $\{1, x, x^2, \dots\}$ in the Hilbert space $L^2(\mathbb{R}, d\mu)$. These polynomials satisfy the orthogonality condition $\int P_n(x) P_m(x) d\mu(x) = \delta_{nm} \|P_n\|^2$.

A characteristic and defining signature of orthogonal polynomials on the real line (OPRL) is the three-term recurrence relation. For orthonormalized polynomials $p_n(x) = P_n(x)/\|P_n\|$, this recurrence takes the form:

$$xp_n(x) = a_{n+1}p_{n+1}(x) + b_{n+1}p_n(x) + a_np_{n-1}(x)$$

where the coefficients a_n are strictly positive and b_n are real. The proof of this relation relies entirely on the orthogonality of the basis. Because $xp_n(x)$ is a polynomial of degree $n + 1$, it can be expanded in the basis as $\sum_{k=0}^{n+1} c_k p_k(x)$. The coefficient c_k is obtained via the inner product $c_k = (xp_n, p_k)$. By the self-adjoint nature of multiplication by a real variable, $(xp_n, p_k) = (p_n, xp_k)$. For $k < n - 1$, xp_k is a polynomial of degree at most $n - 1$, which is orthogonal to p_n , meaning $c_k = 0$. Thus, only the terms for $k = n - 1, n$, and $n + 1$ survive, yielding the three-term recurrence.

This recurrence directly translates to the multiplication operator by x being represented by a tridiagonal, symmetric Jacobi matrix J in the polynomial basis. The spectral measure of this Jacobi matrix is precisely the measure μ . The asymptotic behavior of these recurrence coefficients encodes deep topological and metric properties of the support of the measure.

Rakhmanov's theorem serves as a landmark result in this domain. It establishes a rigorous connection between the macroscopic continuous properties of the measure and the microscopic discrete limits of the recurrence coefficients. Rakhmanov proved that for a measure μ supported on $[-1, 1]$ with an absolutely continuous part whose derivative $\mu'(x) > 0$ almost everywhere on the interval, the recurrence coefficients converge asymptotically to those of the free case (the Chebyshev polynomials). Rigorously, this manifests as $\lim_{n \rightarrow \infty} a_n = 1/2$ and $\lim_{n \rightarrow \infty} b_n = 0$. The Denisov-Rakhmanov theorem extends this beyond simple intervals to operators with essential support equal to $[-2, 2]$. It states that if the absolutely continuous spectrum fills the interval $\Sigma_{ac}(\mu) = [-2, 2]$, then the Jacobi parameters must asymptotically approach the free parameters.

The Christoffel-Darboux Kernel and Spectral Perturbations

To understand the localized behavior of orthogonal polynomials and their associated measures, we invoke the Christoffel-Darboux (CD) kernel, defined as the sum of the outer products of the orthonormal polynomials up to degree n :

$$K_n(x, y) = \sum_{j=0}^n p_j(x)p_j(y)$$

This kernel acts as the reproducing kernel for the finite-dimensional Hilbert space of polynomials of degree at most n . The proof of its reproducing property is immediate from the orthonormality of the basis: for any polynomial f of degree $\leq n$, expanded as $f(x) = \sum_{j=0}^n c_j p_j(x)$, the integral operator satisfies $\int K_n(x, y)f(x)d\mu(x) = \sum_{j=0}^n p_j(y) \int p_j(x)f(x)d\mu(x) = \sum_{j=0}^n c_j p_j(y) = f(y)$.

The reciprocal of the kernel evaluated on the diagonal, $\lambda_n(x) = 1/K_n(x, x)$, is known as the Christoffel function. The Christoffel function provides a strict, mathematically proven upper bound for the maximal mass that any discrete atomic measure, matching the exact moments up to degree $2n$, can place at the point x . The asymptotics of the Christoffel function $n\lambda_n(x)$ as $n \rightarrow \infty$ yield the density of the equilibrium measure of the support, linking the algebraic properties of the polynomials to potential theory.

The classical Christoffel-Darboux formula provides a closed-form expression for the kernel:

$$K_n(x, y) = a_{n+1} \frac{p_{n+1}(x)p_n(y) - p_n(x)p_{n+1}(y)}{x - y}$$

This identity can be proven elegantly using operator commutators. Let M_x be the operator of multiplication by x , and let π_n be the orthogonal projection onto the subspace of polynomials of degree up to n . The commutator $[M_x, \pi_n]$ isolates the boundary terms of the projection. By evaluating the matrix elements of this commutator, one recovers the right-hand side of the CD formula, demonstrating that the kernel's structure is a direct consequence of the tridiagonal nature of the Jacobi operator.

Logarithmic Potential Theory and Equilibrium Measures

When analyzing the asymptotic distribution of the zeros of orthogonal polynomials, algebraic methods give way to the analytic rigor of logarithmic potential theory. For a compact set $E \subset \mathbb{C}$ with positive logarithmic capacity, the electrostatic energy integral for a probability measure μ supported on E is defined as:

$$I[\mu] = \iint \log|z - t|^{-1} d\mu(t) d\mu(z)$$

The equilibrium measure μ_E is the unique probability measure that minimizes this energy integral over all valid unit measures supported on E . The minimal energy defines the capacity of the set via the relation $V = -\log(\text{cap}(E))$.

For the specific interval $E = [-1, 1]$, the optimal distribution that minimizes the electrostatic energy can be derived exactly. The unique solution is the arcsine distribution, defined by the absolutely continuous measure:

$$d\mu_{[-1,1]}(x) = \frac{1}{\pi\sqrt{1-x^2}} dx$$

This distribution possesses an equalization property, a phenomenon where the logarithmic potential is mathematically constant across the entire interval, ensuring that no individual "charge" in the electrostatic analogy has an incentive to migrate.

The connection between this continuous potential theory and discrete polynomial roots is established by the theorem of Jentzsch-Szegő. Extended to analytic curves and compact Riemann surfaces by modern researchers, the theorem dictates that the discrete measure ν_n , formed by placing a Dirac mass of $1/n$ at each of the n zeros of the n -th orthogonal polynomial, converges in the weak-star topology to the continuous equilibrium measure μ_E as $n \rightarrow \infty$. This rigorously proves that the roots of polynomials of best approximation asymptotically distribute according to the arcsine law on $[-1, 1]$, clustering densely near the endpoints of the interval and exhibiting sparsity near the origin.

To further formalize this link, the exact calculations of the moments of the continuous arcsine distribution yield a strict combinatorial sequence. Because the measure is symmetric, all odd moments vanish ($\mu_{2k+1} = 0$). The even moments are calculated via the integral:

$$\mu_{2k} = \int_{-1}^1 \frac{x^{2k}}{\pi\sqrt{1-x^2}} dx$$

By applying the substitution $x = \sin(\theta)$, this transforms into a standard trigonometric integral evaluated via the Beta function, yielding exactly $\mu_{2k} = 4^{-k} \binom{2k}{k}$. This exact mathematical equivalence demonstrates a

profound and unshakeable link between the continuous equilibrium measure of logarithmic potential theory and the discrete combinatorics of the central binomial coefficient.

Szegő Limit Theorems and Toeplitz Determinants

While Hankel matrices arise naturally from moments on the real line, Toeplitz matrices emerge when studying measures defined on the unit circle in the complex plane (OPUC). A Toeplitz matrix T_n is defined by its entries being constant along diagonals, $T_{j,k} = c_{j-k}$, where c_m represents the m -th Fourier coefficient of a real-valued, integrable function $f(\theta)$ defined on the unit circle:

$$c_m = \frac{1}{2\pi} \int_0^{2\pi} e^{-im\theta} f(\theta) d\theta$$

The evaluation of the determinant of large Toeplitz matrices, $D_n(f) = \det(T_n)$, as $n \rightarrow \infty$, is famously resolved by the Szegő limit theorems.

The First Szegő Theorem states that if f is strictly positive and sufficiently smooth, the asymptotic ratio of successive Toeplitz determinants converges to the geometric mean of the function f :

$$\lim_{n \rightarrow \infty} \frac{D_n(f)}{D_{n-1}(f)} = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \log f(\theta) d\theta \right)$$

The proof of this theorem fundamentally relies on the realization that the determinant of the Toeplitz matrix is equivalent to the product of the squared norms of the orthogonal polynomials on the unit circle. The geometric mean of the symbol f represents the entropy of the associated measure, dictating the leading-order exponential growth rate of the determinant.

However, modern spectral theory and statistical mechanics frequently require bounding the exact non-perturbative fluctuation around this leading-order growth. The Strong (or Second) Szegő Limit Theorem establishes this finer bound. By expressing the Fourier coefficients of the logarithm of the symbol $\log f(\theta)$ as $\hat{L}_k$, the strong theorem proves that the ratio of the determinant to its limiting geometric approximation converges to an explicit exponential series involving the Sobolev norm of the logarithm of the symbol:

$$\lim_{n \rightarrow \infty} \frac{D_n(f)}{e^{(n+1)\bar{L}_0}} = \exp \left(\sum_{k=1}^{\infty} k |\hat{L}_k|^2 \right)$$

where the right-hand side is finite provided that f and e^f are in L^1 and the sum converges. Barry Simon and subsequent researchers extended these theorems beyond simple intervals or the entire unit circle to finite gap sets and infinite gap sets of Parreau-Widom type. In these complex geometries, the product of the recurrence coefficients must be normalized by the n -th power of the logarithmic capacity of the set, $\text{Cap}(E)^n$, to achieve a non-trivial limit, proving that capacity and entropy are inextricably linked across all measure spaces.

Algorithmic Stability and Hankel Matrix Ill-Conditioning

While the theoretical properties of Hankel and Toeplitz matrices reveal deep algebraic elegance, their practical application in numerical computation is fraught with severe instability. Positive semidefinite Hankel matrices, which arise naturally from moment sequences of positive measures on intervals, are known to be exponentially ill-conditioned, meaning the ratio of their largest to smallest singular values grows exponentially with the matrix dimension.

Rigorous numerical investigations into the conditioning of $N \times N$ Hankel matrices substantiate this claim. To demonstrate the scale of this instability, we analyze the condition numbers derived from two distinct moment sequences: the continuous arcsine distribution and a discrete uniform distribution defined on integer nodes.

Matrix Size (N×N)	Smallest Singular Value ($\sigma_{\min}$)	Largest Singular Value ($\sigma_{\max}$)	Condition Number (κ)
4×4	1.971×10^{-2}	1.277×10^0	6.480×10^1
6×6	7.199×10^{-4}	1.470×10^0	2.042×10^3
8×8	2.467×10^{-5}	1.629×10^0	6.604×10^4
10×10	8.159×10^{-7}	1.768×10^0	2.167×10^6
Table 1: Singular value decay and exponential growth of the condition number for Hankel matrices constructed from the exact continuous moments of the arcsine distribution on $[-1,1]$. Data confirms the rapid deterioration of numerical stability.			

When the generating measure is discrete and non-normalized, the ill-conditioning becomes catastrophic even at trivial dimensions. For a discrete uniform measure defined on nodes $1, \dots, 30$, the conditioning explodes.

Matrix Size (N×N)	Smallest Singular Value ($\sigma_{\min}$)	Largest Singular Value ($\sigma_{\max}$)	Condition Number (κ)
4×4	4.537×10^{-2}	1.169×10^8	2.576×10^9

Matrix Size (N×N)	Smallest Singular Value ($\sigma_{\min}$)	Largest Singular Value ($\sigma_{\max}$)	Condition Number (κ)
6×6	1.087×10^{-2}	6.415×10^{13}	5.900×10^{15}
8×8	1.593×10^{-1}	5.833×10^{17}	3.662×10^{18}
10×10	2.950×10^{-1}	2.299×10^{18}	7.793×10^{18}
Table 2: Condition numbers for Hankel matrices generated by discrete uniform integer nodes on [1,30]. The matrix exceeds the limits of standard double-precision floating-point arithmetic at $N = 6$.			

Normalizing the discrete nodes to the interval $[-1,1]$ via an affine transformation maps the sequence back into a manageable range, but the exponential decay of the singular values persists, mirroring the continuous arcsine case.

Matrix Size (N×N)	Smallest Singular Value ($\sigma_{\min}$)	Largest Singular Value ($\sigma_{\max}$)	Condition Number (κ)
4×4	1.924×10^{-2}	1.139×10^0	5.922×10^1
6×6	7.685×10^{-4}	1.211×10^0	1.576×10^3
8×8	2.724×10^{-5}	1.261×10^0	4.631×10^4
10×10	8.788×10^{-7}	1.302×10^0	1.482×10^6

Matrix Size (N×N)	Smallest Singular Value ($\sigma_{\min}$)	Largest Singular Value ($\sigma_{\max}$)	Condition Number (κ)
Table 3: Condition numbers for normalized discrete uniform nodes mapped to $[-1,1]$. While scale is preserved, exponential ill-conditioning remains an intrinsic property of the Hankel structure.			

The theoretical justification for this exponential decay was definitively proven by Beckermann and Townsend. They demonstrated that for any positive semidefinite Hankel matrix $H_n \in \mathbb{R}^{n \times n}$, the singular values σ_k decay exponentially with respect to the index k . Specifically, the k -th singular value is bounded by:

$$\sigma_k(H_n) \leq C \rho^{-k/\log n} \|H_n\|_2$$

where $C > 0$ and $\rho > 1$ are explicitly calculable constants, and $\|H_n\|_2$ is the spectral norm.

The proof of the Beckermann-Townsend bound is a masterpiece of complex approximation theory, relying heavily on Zolotarev numbers. A Zolotarev number $Z_k(E, F)$ measures the minimax optimal rational approximation between two disjoint sets E and F in the complex plane, defined as the infimum of the ratio of the supremum of a rational function of degree (k, k) on E to its infimum on F . By factoring the Hankel matrix through an extremal problem involving these rational functions, Beckermann and Townsend proved that the rank of the algebraic difference between scaled matrix operators is bounded by the degrees of the rational approximants. Consequently, any positive semidefinite Hankel matrix can be approximated to an accuracy of $\epsilon \|H_n\|_2$ by a matrix of rank $O(\log n \log(1/\epsilon))$.

This theoretical bound has profound algorithmic implications. Because positive semidefinite Hankel matrices are inherently low-rank, full $O(n^3)$ factorizations are computationally wasteful and numerically catastrophic. Instead, sublinear time algorithms utilizing randomized pivoted Cholesky decomposition can be deployed. These algorithms evaluate the diagonal elements and sequentially compute a low-rank factorization BB^* in $O(n \cdot \text{polylog}(n, 1/\epsilon))$ time. The pivoted approach guarantees that the error perfectly matches the Beckermann-Townsend theoretical limits without ever materializing the full, ill-conditioned matrix in memory.

Such structural factorizations also allow for rapid transformations between orthogonal polynomial bases. The conversion matrix between Chebyshev and Legendre polynomial coefficients can be analytically decomposed into a diagonally scaled Hadamard product of an upper-triangular Toeplitz matrix and a severely ill-conditioned Hankel matrix, $M = D_1(T \circ H)D_2$. By applying the low-rank Cholesky approximation to the Hankel component H , the basis conversion is executed in $O(N \log^2 N)$ operations via the Fast Fourier Transform (FFT), avoiding the $O(N^2)$ barrier of direct transformation.

Combinatorial Asymptotics and Probabilistic Limits

The exact evaluation of continuous moments, such as the arcsine moments $\binom{2n}{n}4^{-n}$, requires robust discrete asymptotic expansions to bridge continuous operator theory with discrete algorithms. The foundational tool for such expansions is Stirling's approximation for the gamma function. Derived rigorously via the saddle point method (a multidimensional extension of Watson's lemma), the logarithm of the gamma function possesses the formal asymptotic expansion:

$$\ln\Gamma(z) \sim \left(z - \frac{1}{2}\right)\ln z - z + \frac{1}{2}\ln(2\pi) + \sum_{k=1}^{\infty} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}$$

where B_{2k} represent the Bernoulli numbers ($B_2 = 1/6, B_4 = -1/30, B_6 = 1/42$, etc.). The saddle point method deforms the contour of integration for the Laplace transform of the gamma function to pass through the stationary point of the phase, proving that the local Gaussian integral dominates the asymptotic behavior as $|z| \rightarrow \infty$.

Applying this expansion to the ratio defining the normalized central binomial coefficient, $\Gamma(n + 1/2)/(\sqrt{\pi}\Gamma(n + 1))$, yields a precise Taylor expansion in terms of $u = 1/n$. The asymptotic series diverges from the simple geometric 4^n scaling, exhibiting higher-order polynomial corrections required for precision numerical approximations:

$$\ln\Gamma(z) \sim \left(z - \frac{1}{2}\right)\ln z - z + \frac{1}{2}\ln(2\pi) + \sum_{k=1}^{\infty} \frac{B_{2k}}{2k(2k-1)z^{2k-1}}$$

This formula provides the exact boundary behavior for the moments of the arcsine distribution as $n \rightarrow \infty$.

The systematic generation of these asymptotic series relies on the composition and inversion of generating functions. While partial Bell polynomials are traditionally used, De Moivre polynomials (also known as partial ordinary Bell polynomials) provide a more natural algebraic ring for these formal power series transformations. Defined by the expansion of the k -th power of a power series $A(t) = \sum a_j t^j$, the De Moivre polynomials $A_{n,k}(a_1, a_2, \dots)$ explicitly encode the combinatorial sums over integer partitions. This allows for the algorithmic generation of high-order asymptotic corrections, such as those required for extended binomial distributions, without the computational burden of recursive analytical derivative evaluations.

This rigorous combinatorial tracking extends to probability theory, specifically the Central Limit Theorem (CLT) for discrete independent trials. When $S_n = X_1 + \dots + X_n$ is a sum of discrete random variables drawn from a uniform distribution with finite variance, the standardized sum converges to a continuous normal distribution. The error bounds of this approximation are governed by the local limit theorem. By constructing a spike graph of the discrete states, the theorem maps the discrete probability impulses directly onto the standard normal density $\phi(x)$ using standard continuity corrections. Thus, the macroscopic continuous variables observed in equilibrium measures are proven to emerge strictly from the aggregate limit of discrete combinatorial events, unifying probability theory with potential theory.

Contraction Mappings in Iterative Linear Solvers

The solution of linear systems governed by structured matrices often relies on iterative refinement when direct inversion is precluded by the aforementioned conditioning issues. The Jacobi method transforms a strictly diagonally dominant linear system $Ax = b$ into a fixed-point iteration by decomposing the matrix as $A = D + M$, where D is the diagonal component and M contains the off-diagonal elements. The iteration is defined by the affine transformation:

$$x^{(k+1)} = D^{-1}b - D^{-1}Mx^{(k)} = c - Jx^{(k)}$$

where $J = D^{-1}M$ is defined as the Jacobi iteration matrix.

The convergence of this iterative algorithm relies entirely on the spectral radius of the matrix J , denoted $\rho(J)$. If the magnitude of the largest eigenvalue of J is strictly less than 1 ($\rho(J) < 1$), the affine transformation $f(x) = c - Jx$ is rigorously proven to be a contraction mapping. By the Banach fixed-point theorem, the distance between successive approximations strictly shrinks in the associated metric space, guaranteeing convergence to the unique global solution independent of the choice of the initial vector $x^{(0)}$. The geometric interpretation of this contraction provides a fundamental model for stability in dynamical systems and recursive constraint propagation.

A Rigorous Mathematical Audit of Meta-Computational Paradigms (The Nexus Framework)

In stark contrast to the rigorously established fields of potential theory, spectral analysis, and numerical linear algebra—where every claim is backed by formal proofs and limit theorems—certain contemporary literature proposes sweeping meta-computational ontologies to explain physical phenomena. One such paradigm is the "Nexus Framework," driven by Dean Kulik. The framework asserts an "Ontological Inversion," claiming that reality does not run *on* a computational substrate, but rather *is* the computational substrate. It classifies physical entities as "frozen verbs," replacing continuous geometries with discrete recursive operations.

Applying the strict laws of mathematical inference and the absolute mandate for rigorous proof, a thorough audit of the Nexus Framework reveals it to be a philosophical heuristic entirely lacking in formal mathematical validity. The following core claims are scrutinized and falsified:

1. The Universal Harmonic Attractor ($H = 0.35$)

The framework posits a "Mark 1 Harmonic Constant" of $H = 0.35$, asserting that the universe strictly allocates exactly 35% of its processing power to "actualized structure" and 65% to uncollapsed potential. This is claimed to serve as a universal tuning fork for systemic criticality. **Audit Result:** This claim is mathematically vacuous and dimensionally arbitrary. In theoretical physics and dynamical systems, critical transitions (e.g., phase transitions, self-organized criticality, or renormalization group fixed points) are governed by dimensionless constants derived explicitly from symmetric field equations, partition functions, or topological invariants. There is no variational principle, Hamiltonian formalism, or measure-theoretic proof provided to derive the exact integer ratio 35/100 as a bounding constraint on any physical state space. The assignment of a fixed percentage to abstract "processing power" conflates discrete algorithmic resource allocation with continuous energy conservation without establishing a formal mathematical isomorphism.

2. Pi (π) and SHA-256 as Deterministic Trust Lattices

The framework claims that the transcendental number π is not irrational in a "random sense" but serves as a deterministic "checksum lattice" and "address space" for universal harmony, utilizing algorithms like the Bailey–Borwein–Plouffe (BBP) formula as a "read-head". It further attempts to map physical wave-function collapse to the topological geometry of cryptographic hashing like SHA-256. **Audit Result:** This represents a fundamental category error spanning number theory, cryptography, and physics. The BBP formula allows for the extraction of the n -th hexadecimal digit of π without computing preceding digits strictly due to specific modular algebraic properties of hypergeometric series, not because π enforces a background

physical spatial coherence. Furthermore, π is widely conjectured to be a normal number, meaning its digit sequences are asymptotically uniformly distributed; thus, any finite sub-pattern can be found within it by sheer statistical inevitability. Calling it a "checksum" is algebraically meaningless, as checksums require finite cyclic groups (e.g., Galois fields) to detect bit-level mutations, whereas π is an infinite, non-repeating continuous scalar.

Similarly, mapping SHA-256—a highly non-linear, discrete, non-invertible bitwise permutation over finite fields designed explicitly to destroy structural topology—to the continuous symmetry groups (such as the $SU(3) \times SU(2) \times U(1)$ Lie groups) that dictate physical conservation laws violates Noether's Theorem. Noether's Theorem rigorously proves that conservation laws arise strictly from continuous symmetries of the action, which cannot exist in a discrete hash permutation.

3. Zero-Point Harmonic Collapse and the Typeless Universe

The Nexus model relies on "Zero-Point Harmonic Collapse" to explain quantum measurement and phase shifts, rejecting object-oriented physics for a "Typeless Universe" where objects are merely cyclic loops of recursive operations. **Audit Result:** While the theory of recursive feedback is valid in non-linear dynamics, the outright rejection of "static noun" objects (mass, charge, spin) contradicts the foundational representation theory of Lie algebras. Elementary particles are rigorously proven to be the irreducible representations of the Poincaré group, a foundational mathematical reality that cannot be waved away by declaring them "verbs." Furthermore, as demonstrated by the Szegő limit theorems and the Denisov-Rakhmanov theorem discussed earlier, discrete recursive parameter spaces (like Jacobi and Verblunsky coefficients) map exactly and uniquely to continuous spectral measures. Attempting to override the continuous topologies of General Relativity and Quantum Mechanics with a discrete "typeless" algebra requires, at minimum, a proof of convergence in the continuum limit (such as the convergence of discrete orthogonal polynomials to continuous equilibrium measures via Jentzsch-Szegő). The Nexus framework provides no such limit theorems, rendering its "Collapse Signature Theory" algebraically unprovable and physically inert.

Conclusion

The rigorous mathematical analysis of orthogonal polynomials, moment problems, and their structured matrix representations reveals a profound and provable hierarchy linking discrete sequences to continuous macroscopic phenomena. The necessary and sufficient conditions for the existence of moment measures rest firmly on the proven positive semidefiniteness of Hankel matrices, while their uniqueness is determined by the asymptotic divergence of the moments as dictated by Carleman's criterion. In the complex plane, the Szegő limit theorems provide exact geometric and entropic bounds for the determinants of Toeplitz matrices, governing the spectral behavior of systems defined on the unit circle.

Simultaneously, the inherent numerical instability of these discrete structures—demonstrated by the exponential decay of singular values in positive semidefinite Hankel matrices—forces the adoption of advanced, low-rank rational approximation techniques. The bounds provided by Zolotarev numbers and sublinear Cholesky factorizations solve these computational bottlenecks without sacrificing mathematical rigor. This entire architecture is unified by formal proofs, bridging logarithmic potential theory, the arcsine distribution, and the explicit combinatorial expansions of Stirling and De Moivre.

When contrasted with heuristic meta-computational philosophies, the necessity of rigorous inference becomes paramount. While conceptual models may propose unifying digital substrates, only those

frameworks grounded in proven limit theorems, capacity-minimizing energy integrals, and convergent operator topologies can be admitted into the canonical structure of theoretical physics and computational mathematics. The mathematical universe requires no metaphysical interpretation; its truths are wholly encapsulated within the rigorous bounds of its limits, recurrences, and spectral measures.

Works cited

1. Moment problem - Wikipedia, https://en.wikipedia.org/wiki/Moment_problem
2. The moment problem, https://web.williams.edu/Mathematics/sjmiller/public_html/book/papers/jcmp.pdf
3. The Classical Moment Problem | ari1110 - note, <https://note.com/ari1110/n/nd7b70d57c460?hl=en>
4. The Problem of Moments: A Bunch of Classical Results with Some Novelties - MDPI, <https://www.mdpi.com/2073-8994/15/9/1743>
5. 3.2 Jacobi Polynomials 3.2.1 Basic Properties - LSEC, <https://lsec.cc.ac.cn/~hyu/teaching/shonm2013/STWchap3.2p.pdf>
6. Recurrence coefficients for the time-evolved Jacobi weight and discrete Painlevé equations on the S_{D_5} Sakai surface - arXiv, <https://arxiv.org/pdf/2511.04176>
7. arXiv:math/0702388v2 [math.SP] 6 Dec 2008, <https://arxiv.org/pdf/math/0702388>
8. the analytic theory of matrix orthogonal polynomials - Mathematical Physics Preprint Archive mp_arc, http://kleine.mat.uniroma3.it/mp_arc/c/07/07-278.pdf
9. Orthogonal Polynomials and Approximation Theory: some open problems, <https://wis.kuleuven.be/analyse/members/walter/jatopen-problems.pdf>
10. Pointwise Asymptotics for Orthonormal Polynomials at the Endpoints of the Interval via Universality | International Mathematics Research Notices | Oxford Academic, <https://academic.oup.com/imrn/article/2020/4/961/4952086>
11. Asymptotics of Orthogonal Polynomials: Some Old, Some New, Some Identities, https://csclub.uwaterloo.ca/~pbarfuss/Lubinsky2000_AsymptoticsOfOrthogonalPolynomials.pdf
12. The Christoffel–Darboux Kernel - UT Math, https://web.ma.utexas.edu/mp_arc/c/08/08-107.pdf
13. Perturbations of Christoffel–Darboux Kernels: Detection of Outliers - Department of Mathematics, <http://www.math.vanderbilt.edu/saffeb/texts/275.pdf>
14. of best approximants - h.-p. blatt, eb saff and m. simkani - Department of Mathematics, <http://www.math.vanderbilt.edu/saffeb/texts/89.pdf>
15. Logarithmic Potential Theory with Applications to Approximation Theory - ResearchGate, https://www.researchgate.net/publication/47451593_Logarithmic_Potential_Theory_with_Applications_to_Approximation_Theory

16. [unknown_url](#)
17. Greed, hedging, and acceleration in convex optimization - DSpace@MIT, <https://dspace.mit.edu/bitstreams/d482e1ec-d62b-4eeb-a233-cca937edo666/download>
18. THE THEOREM OF JENTZSCH–SZEGŐ ON AN ANALYTIC CURVE: APPLICATION TO THE IRREDUCIBILITY OF TRUNCATIONS OF POWER SERIES | International Journal of Number Theory, <https://www.worldscientific.com/doi/10.1142/S1793042111004691>
19. In memoriam: Herbert Stahl August 3, 1942–April 22, 2013, <https://history-of-approximation-theory.com/fpapers/stahl.pdf>
20. Szegő limit theorems - Wikipedia, https://en.wikipedia.org/wiki/Szeg%C5%91_limit_theorems
21. Szegő's Theorem and Its Descendants: Spectral Theory for L^2 Perturbations of Orthogonal Polynomials [Course Book ed.] 9781400837052, <https://dokumen.pub/szegos-theorem-and-its-descendants-spectral-theory-for-ltsupgt2lt-supgt-perturbations-of-orthogonal-polynomials-course-booknbsped-9781400837052.html>
22. On Szegő's limit theorem - ResearchGate, https://www.researchgate.net/publication/231116223_On_Szego's_limit_theorem
23. Szegő's theorem on Parreau–Widom sets - arXiv, <https://arxiv.org/pdf/1104.1915>
24. Barry Simon Books - MacTutor History of Mathematics - University of St Andrews, https://mathshistory.st-andrews.ac.uk/Extras/Simon_Barry_books/
25. Sublinear Time Low-Rank Approximation of Hankel Matrices - arXiv, <https://arxiv.org/pdf/2511.21418>
26. arXiv:1609.09494v1 [math.NA] 29 Sep 2016, <https://arxiv.org/pdf/1609.09494>
27. Bounds on the Singular Values of Matrices with Displacement Structure - SIAM.org, <https://epubs.siam.org/doi/10.1137/19M1244433>
28. fast polynomial transformations based on Toeplitz and Hankel matrices - The University of Manchester, <https://personalpages.manchester.ac.uk/staff/marcus.webb/pdfs/webbfastconversion.pdf>
29. On Symmetric Factorizations of Hankel Matrices - arXiv, <https://arxiv.org/pdf/2307.00805>
30. Spectral Methods for Polynomial Optimization and Adaptive Sampling in Kernel Quadratur, https://thesis.caltech.edu/18695/01/PhD_Thesis.pdf
31. arXiv:1604.07486v2 [math.NA] 27 Apr 2016, <https://arxiv.org/pdf/1604.07486>
32. Stirling's approximation - Wikipedia, https://en.wikipedia.org/wiki/Stirling%27s_approximation

33. (PDF) Asymptotic approximations of integrals: An introduction, with recent developments and applications to orthogonal polynomials - ResearchGate,
https://www.researchgate.net/publication/265871043_Asymptotic_approximations_of_integrals_An_introduction_with_recent_developments_and_applications_to_orthogonal_polynomials
34. Asymptotic Expansions of Central Binomial Coefficients and Catalan Numbers,
<https://cs.uwaterloo.ca/journals/JIS/VOL17/Elezovic/elezovic5.pdf>
35. Stirling's Approximation for Central Extended Binomial Coefficients - arXiv,
<https://arxiv.org/pdf/1203.2122>
36. asymptotics of central binomial coefficient - Planetmath,
<https://planetmath.org/asymptoticsofcentralbinomialcoefficient>
37. De Moivre and Bell polynomials - CUNY,
<https://fsw01.bcc.cuny.edu/cormac.osullivan/Research/demoivre-paper.pdf>
38. 9.2: Central Limit Theorem for Discrete Independent Trials - Statistics LibreTexts,
[https://stats.libretexts.org/Bookshelves/Probability_Theory/Introductory_Probability_\(Grinstead_and_Snell\)/09%3A_Central_Limit_Theorem/9.02%3A_Central_Limit_Theorem_for_Discrete_Independent_Trials](https://stats.libretexts.org/Bookshelves/Probability_Theory/Introductory_Probability_(Grinstead_and_Snell)/09%3A_Central_Limit_Theorem/9.02%3A_Central_Limit_Theorem_for_Discrete_Independent_Trials)
39. [1203.2122] Stirling's approximation for central extended binomial coefficients - arXiv,
<https://arxiv.org/abs/1203.2122>
40. Jacobi method - Wikipedia, https://en.wikipedia.org/wiki/Jacobi_method
41. The geometry of Jacobi's method - The DO Loop - SAS Blogs,
<https://blogs.sas.com/content/iml/2024/01/29/geometry-jacobi-method.html>
42. 11.1 Jacobi Transformations of a Symmetric Matrix - IATE,
<https://iate.oac.uncor.edu/~mario/materia/nr/numrec/f11-1.pdf>
43. Linear Algebra Algorithms as Dynamical Systems - Moody T. Chu,
https://mtchu.math.ncsu.edu/Research/Papers/la_dao2.pdf
44. (PDF) The Nexus Framework: A Unified Meta-Computational Ontology of Recursive Harmonic Folding - ResearchGate,
https://www.researchgate.net/publication/401306736_The_Nexus_Framework_A_Unified_Meta-Computational_Ontology_of_Recursive_Harmonic_Folding
45. (PDF) The Nexus Framework: An Exhaustive Operational Manual of Recursive Harmonic Formulas and Substrate Architecture - ResearchGate,
https://www.researchgate.net/publication/401144769_The_Nexus_Framework_An_Exhaustive_Operational_Manual_of_Recursive_Harmonic_Formulas_and_Substrate_Architecture
46. The Nexus Recursive Harmonic Framework: A Meta-Computational Ontology of Spacetime, Biology, and Cryptographic Geometry - DOI, <https://doi.org/10.5281/zenodo.19211659>

47. Dean KULIK | Developer | Research and Development - ResearchGate,
<https://www.researchgate.net/profile/Dean-Kulik>
48. The Nexus Framework: A Comprehensive Analysis of its Recursive Harmonic Principles and Unifying Potential - Zenodo, <https://zenodo.org/records/15903358>
49. The Mark1 Nexus: A Recursive System Treatise - OpenAIRE - Explore,
<https://explore.openaire.eu/search/result?pid=10.5281/zenodo.15871553>